Discrete Mathematics Spring 2026

Homework 3: Inference and First Order Logic (FOL)

Due Sunday February 15 @11:59:00pm Eastern

Show all your work to receive full credit

1. **Validity**

   Consider the following inference rule.

   $$\frac{\text{I am cold} \rightarrow \text{I forgot my jacket} \qquad \text{I forgot my jacket}}{\text{I am cold}}$$

   (a) Is this inference rule valid?

   (b) Briefly explain your answer.

   (c) Use a truth table to verify your answer by checking whether the formula $((p \rightarrow q) \wedge q) \rightarrow p$ is a tautology, where p represents "I am cold" and q represents "I forgot my jacket".

2. **Inference**

   Consider the following premises:

   1. rainy $\rightarrow$ (cold $\vee$ windy)
   2. $\neg$cold
   3. windy $\rightarrow$ uncomfortable
   4. $\neg$uncomfortable
   5. stormy $\rightarrow$ rainy

   Prove that we can derive the conclusion:

   $$\neg\text{stormy}$$

   Make sure to cite the inference rule/PL law you used at each step. Any format you use (steps, inference lines, etc.) will be accepted as far as it is legible and clear.

3. **FOL Propositions**

   Let the domain be $B =$ "Books". Consider the following propositions:

   1. All bestselling novels are engaging and well-written.
   2. Some textbooks are not difficult.
   3. All biographies are informative, and some novels are bestselling.

   Translate the above sentences to first order logic (FOL). Use only the predicates: novel(x), bestselling(x), engaging(x), wellwritten(x), textbook(x), difficult(x), biography(x), informative(x).

4. **FOL Propositions**

   Let the domain be $P =$ "People" and $M =$ "Movies". Translate the following English sentences into first-order logic using the predicates: comedy(y), drama(y), enjoys(x, y), directed(x, y), acclaimed(y), where $x \in P$ represents a person and $y \in M$ represents a movie.

   (a) Every person enjoys at least one comedy.

   (b) There is a drama that all people enjoy.

(c) No person enjoys every comedy.

(d) If a person directed a movie, then that person enjoys that movie.

(e) Every acclaimed movie is directed by some person.

5. **Negation of FOL Propositions**

For each of the following FOL formulas, write an equivalent formula by pushing negations inward (using quantifier negation rules and De Morgan's laws) until each negation symbol appears directly before a predicate.

(a) $\neg(\forall x\ (P(x) \rightarrow Q(x)))$

(b) $\neg(\exists x\ (R(x) \wedge \forall y\ S(x, y)))$

(c) $\neg(\forall x\ (\exists y\ P(x, y) \rightarrow \forall z\ Q(z)))$

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